

DEBRECEN, Hungary

WMO: 128820

Lat: 47.489N

Lon: 21.615E

Elev: 110

StdP: 100.01

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-9.2	-15.2	1.0	-9.3	-12.8	1.3	-7.5	9.9	0.7	9.1	-0.5	2.1	50	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	32.8	21.2	31.1	20.6	29.5	20.0	22.6	29.9	21.7	28.7	20.9	27.6	2.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	15.1	25.8	19.3	14.2	24.9	18.5	13.5	24.0	67.0	30.1	63.6	29.0	60.7	27.5	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.5	6.7	-15.1	34.8	3.4	1.4	-17.5	35.8	-19.5	36.7	-21.4	37.5	-23.8	38.6		
(5)			-15.5	24.1	3.3	0.9	-17.8	24.7	-19.8	25.2	-21.6	25.7	-24.0	26.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-0.7	1.3	5.9	12.0	16.7	20.1	21.8	21.7	16.7	11.2	5.9	0.4		
	DBStd	8.99	4.59	4.84	4.41	3.95	3.54	3.37	3.04	3.14	3.57	4.15	4.46	4.64		
	HDD10.0	1212	333	242	138	26	1	0	0	0	1	37	137	298		
	HDD18.3	2978	591	476	386	193	74	20	6	8	73	222	373	556		
	CDD10.0	1623	0	0	11	84	209	303	366	362	201	73	13	1		
	CDD18.3	348	0	0	0	1	24	73	113	111	24	1	0	0		
	CDH23.3	3466	0	0	0	27	266	703	1104	1107	242	16	0	0		
	CDH26.7	1186	0	0	0	3	53	226	412	431	62	0	0	0		
(14) Wind	WSAvg	2.9	3.0	3.2	3.6	3.3	3.1	2.9	2.7	2.6	2.7	2.6	2.8	2.9		
(15) Precipitation	PrecAvg	559	31	32	33	45	60	73	66	52	42	37	45	42		
	PrecMax	846	84	117	127	162	161	176	178	124	122	145	116	104		
	PrecMin	321	5	0	2	5	7	19	6	8	0	1	0	0		
	PrecStd	112	17	22	23	26	37	37	40	30	31	31	26	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	15.2	20.1	26.2	30.2	33.0	34.9	35.2	31.6	25.2	19.3	12.3		
		MCWB	8.2	10.2	11.9	16.2	18.6	22.1	21.3	20.9	19.2	17.1	14.0	9.8		
	2%	DB	8.9	12.0	17.6	23.6	28.2	31.1	32.5	33.1	28.3	23.1	16.7	9.9		
		MCWB	7.2	8.4	10.4	14.4	18.0	21.1	21.4	20.9	18.0	15.7	12.5	8.2		
	5%	DB	7.0	9.9	15.2	21.6	26.4	29.4	30.7	31.1	26.2	20.7	14.4	7.9		
		MCWB	5.6	7.0	9.4	13.4	17.4	20.1	20.8	20.5	17.4	14.9	11.2	6.6		
	10%	DB	5.4	7.9	13.0	19.6	24.4	27.5	29.0	29.1	23.9	18.2	12.4	6.4		
		MCWB	4.4	5.8	8.4	12.3	16.5	19.4	20.2	20.1	16.6	13.8	10.2	5.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	10.6	12.9	16.8	20.1	23.5	23.7	23.7	20.3	17.9	14.5	10.0		
		MCDB	10.4	14.1	18.1	24.4	26.8	30.3	30.2	31.3	27.7	23.0	18.2	11.9		
	2%	WB	7.3	8.8	11.3	15.3	19.0	22.2	22.5	22.4	19.2	16.5	13.1	8.2		
		MCDB	8.7	11.6	15.9	21.8	25.8	29.0	29.8	29.7	26.1	21.7	15.9	9.7		
	5%	WB	5.8	7.4	10.1	14.2	18.1	21.1	21.7	21.3	18.2	15.4	11.7	6.8		
		MCDB	6.8	9.3	14.1	20.0	24.8	27.5	28.8	28.5	24.4	19.7	13.9	7.7		
	10%	WB	4.4	6.1	8.8	13.1	17.2	20.1	20.9	20.5	17.2	14.2	10.3	5.5		
		MCDB	5.3	7.7	12.7	18.3	23.2	26.1	27.5	27.5	22.5	17.8	12.3	6.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	7.3	9.4	11.0	11.1	10.9	11.3	11.8	10.9	10.4	7.7	5.2		
		MCDBR	7.1	10.1	13.1	14.0	14.1	13.4	14.0	14.7	14.4	13.7	10.8	7.0		
		MCWBR	5.6	6.9	7.5	6.8	6.0	5.5	5.2	5.3	5.8	7.3	7.0	5.5		
	5% WB	MCDBR	6.8	9.1	11.7	12.6	12.6	12.2	12.7	13.0	12.5	12.1	9.6	6.4		
		MCWBR	5.5	6.5	7.2	6.4	5.9	5.5	5.3	5.4	5.5	6.9	6.4	5.2		
(40) Clear-Sky Solar Irradiance	taub		0.333	0.358	0.396	0.450	0.433	0.434	0.453	0.450	0.417	0.386	0.360	0.333		
	taud		2.322	2.282	2.213	2.085	2.201	2.226	2.181	2.187	2.244	2.310	2.329	2.346		
	Ebn at Noon		728	790	813	801	834	833	810	793	782	746	686	679		
	Edh at Noon		75	96	120	150	139	136	141	134	116	93	75	66		
(44) All-Sky Solar Radiation	RadAvg		1.00	1.85	3.16	4.58	5.59	6.03	5.90	5.37	3.75	2.41	1.30	0.77		
	RadStd		0.12	0.30	0.42	0.43	0.53	0.38	0.50	0.47	0.42	0.27	0.13	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	-124	N/A	+30	
(47) Regional (52 neighbors)	+0.66	N/A	N/A	+1.00	+0.47	N/A	-125	-190	+110	+55	

Nomenclature: See separate page